DATA SCIENCE PROJECT REPORT

Data Augmentation using Convolutional Neural Networks (CNN)

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Executive Summary

This project demonstrates image classification using a Convolutional Neural Network (CNN) enhanced with data augmentation techniques. Data augmentation artificially increases the diversity of training images, helping reduce overfitting and improve model generalization.

Project Detail	Description
Dataset	Image Dataset
Technique	Data Augmentation
Model	Convolutional Neural Network (CNN)
Framework	TensorFlow / Keras
Language	Python

Introduction

Data augmentation generates modified versions of existing images through transformations such as rotation, zoom, flipping, shifting, and shearing. These techniques improve the robustness of CNN models.

Dataset Overview

The notebook trains a CNN on the Image Dataset dataset after preprocessing and applying augmentation to the training images.

Code Explanation

The notebook loads the dataset, preprocesses and normalizes the images, configures an ImageDataGenerator with augmentation parameters, builds a CNN model, trains it on augmented batches, evaluates performance, and makes predictions on unseen data.

Benefits of Data Augmentation

• Reduces overfitting • Improves generalization • Increases effective training data • Produces more robust image classifiers

Applications

Medical imaging, autonomous vehicles, facial recognition, quality inspection, retail image search, satellite image analysis, and document recognition.

Future Scope

Explore advanced augmentation methods such as MixUp, CutMix, AutoAugment, transfer learning, and fine-tuning pre-trained CNN architectures.

Conclusion

The project demonstrates how data augmentation enhances CNN performance by increasing dataset diversity and improving model generalization.